

A systems identification approach using Bayes factors to deconstruct the brain bases of emotion regulation

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Cognitive reappraisal is fundamental to cognitive therapies and everyday emotion regulation. Analyses using Bayes factors and an axiomatic systems identification approach identified four reappraisal-related components encompassing distributed neural activity patterns across two independent functional magnetic resonance imaging (fMRI) studies ($n = 182$ and $n = 176$): (1) an anterior prefrontal system selectively involved in cognitive reappraisal; (2) a fronto-parietal-insular system engaged by both reappraisal and emotion generation, demonstrating a general role in appraisal; (3) a largely subcortical system activated during negative emotion generation but unaffected by reappraisal, including amygdala, hypothalamus and periaqueductal gray; and (4) a posterior cortical system of negative emotion-related regions downregulated by reappraisal. These systems covaried with individual differences in reappraisal success and were differentially related to neurotransmitter binding maps, implicating cannabinoid and serotonin systems in reappraisal. These findings challenge ‘limbic’-centric models of reappraisal and provide new systems-level targets for assessing and enhancing emotion regulation.

Emotion regulation is a fundamental process providing contextual and goal-directed influences on feelings, decision-making and behavior. It is crucial for everyday life and for mental and physical health¹. Dysfunctional regulation is associated with a variety of mental health and substance use disorders² as well as risk for chronic physical health conditions. One of the most effective regulatory strategies is reappraisal, which involves reinterpreting the meaning of stimuli and events in service of a regulatory goal (for example, reducing negative emotion). Reinterpretations can include altering construals of actions, causal attributions of intention³, self-identification and distance and prospectives about future consequences⁴, among other cognitive alterations⁵. Reappraisal is a core therapeutic mechanism in cognitive behavioral therapy and related treatments⁶.

Understanding the brain systems underlying emotion regulation allows comparisons with systems involved in other forms of

cognitive control⁷ and those affected by drugs⁸, brain damage, trauma and affective processes⁹, helping to understand how such factors affect regulatory control. Identifying reappraisal-related systems also provides new neurological targets for cognitive behavioral therapy, drug treatments, neurostimulation and other interventions. The major strategy for identifying reappraisal-related brain systems to date involves comparing responses to emotional stimuli with and without a regulatory goal (for example, ref. 7). Regions more active during regulation are typically thought of as generators of reappraisal-related processes, whereas those with decreased activity are labeled as regulation targets. Research using this approach has found that reappraisal activates several brain regions, including the lateral prefrontal cortex, superior temporal cortex and supplementary motor area (SMA), with decreases in ‘target’ regions principally in the amygdala^{10,11}.

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Several outstanding questions raised by these studies motivate the present work. The first concerns which emotion-related regions are affected by reappraisal. The amygdala is the region most consistently downregulated by reappraisal of negative affect¹², but studies and meta-analyses vary on whether amygdala activity is reduced by reappraisal¹³. In addition, many other key affective regions do not appear to be consistently influenced by reappraisal, including the periaqueductal gray (PAG)^{14,15}, hypothalamus, ventral striatum and pallidum¹⁶. Other cortical areas involved in affect generation also do not appear to be consistently downregulated, and some are activated during reappraisal, including ventrolateral prefrontal cortex (vlPFC¹⁷), anterior insula¹⁸, anterior midcingulate¹⁹ and orbitofrontal cortex²⁰. The paucity of downregulation findings could be a power issue related to the small sample sizes typical of previous studies or could, alternatively, reflect a verum lack of influence, placing boundary conditions on the affective processes affected by reappraisal. In addition, affect generation is now thought to include many more regions (and distributed systems) than was appreciated in early emotion regulation studies^{21,22}, and affect generation-related regions across the brain have not been systematically identified and tested for reappraisal effects. A broader approach that identifies areas engaged during negative emotion and then assesses which of these are downregulated and which are not is needed.

Another complication is that both emotion generation and reappraisal are thought to rely heavily on cognitive appraisal²³, which is important for the construction of emotion²⁴. Reappraisal may, thus, involve many of the same processes used to generate emotions but co-opted in service of regulatory goals. The 'generators' activated during reappraisal in previous studies may be specifically engaged during reappraisal, or they could be activated during negative emotion generation as well, signaling a role in common appraisal processes shared across emotion generation and regulation. Reappraisal-selective regions, should they exist, are likely associated with regulatory goals and may provide a key to understanding and assessing regulatory function in the brain. If reappraisal-selective regions can be identified, they could be measured to assess engagement in reappraisal, detect vulnerabilities and provide more specific brain targets for treatments.

Addressing these questions requires moving beyond traditional statistical hypothesis testing to evaluate evidence both for and against effects across multiple contrasts. For example, evaluating whether negative emotion-related regions are downregulated by reappraisal requires identifying emotion-relevant regions (for example, those activated by negative versus neutral images) and then testing evidence for and against reappraisal-related downregulation (for example, reductions with reappraisal versus passive viewing of negative images). Identifying reappraisal-specific regions involves establishing both a positive response to reappraisal and no difference (a null effect) when generating negative emotion without reappraisal. We refer to this as a systems identification approach, modeled on the axiomatic approach of Rutledge and Glimcher²⁵. This approach takes the following logical form: any reappraisal-specific region (1) must respond more strongly when reappraising negative images than when merely looking at them and (2) must not respond differentially to negative versus neutral images in the absence of reappraisal demand. Such regions can be said to track a hypothetical cognitive process that responds selectively during engagement of regulatory goals. If brain regions with these properties can be identified, support is provided that such a process is a meaningful, identifiable neural component of emotion regulation. Evidence for and against each proposition can be assessed with Bayes factors (BFs)²⁶. However, establishing evidence in favor of a null effect requires substantially larger samples than have typically been available²⁷ (for example, $n \geq 120$ to find a ratio of 10:1 in favor of the null; Supplementary Fig. 1).

In this work, we included two individual functional magnetic resonance imaging (fMRI) datasets with large sample sizes ($n = 176$ and $n = 182$), focusing on findings replicated across both studies to improve

generalizability. Participants in both studies viewed aversive images without reappraising ('Look negative'), reappraised a matched set of aversive images ('Regulate negative') and looked at neutral images ('Look neutral') during scanning, as shown in Fig. 1a. We analyzed contrasts related to emotion generation ('Look negative' – 'Look neutral') and reappraisal ('Regulate negative' – 'Look negative') and calculated BFs using a computationally efficient method²⁸ that allowed efficient calculation for each brain voxel²⁹. Combinations of positive and null findings were used to identify brain regions satisfying each of four potential system components (Fig. 1b,c): (1) 'Reappraisal only' regions responding only to reappraisal demand, not negative images; (2) 'Common appraisal' regions activated by negative images and further increased during reappraisal; (3) 'Non-modifiable emotion generation' regions activated by negative images but unaffected by reappraisal; and (4) 'Modifiable emotion generation' regions activated by negative images and reduced by reappraisal (Fig. 1b,c). We identified regions matching each component with 10:1 evidence in favor of the required effect ('strong' evidence, which also satisfies false discovery rate (FDR) $q < 0.05$ correction for multiple comparisons in all cases) and additionally replicated the relevant effects in both studies independently. We then correlated average activity in each system component with individual differences in reappraisal success, defined as the reduction in negative affect for 'Regulate negative' – 'Look negative'. In addition, the functional and neurochemical attributes of these system components were annotated by spatial (map-wise) associations with Neurosynth topic maps derived from 11,406 previous studies³⁰ and positron emission tomography (PET)-derived molecular imaging maps from 18 neurotransmitter receptors and transporters³¹.

Results

Behavioral results

The three major reappraisal strategies participants used, from those defined by McRae et al.⁵, were 'Change current circumstances', 'Reality challenge' and 'Change future consequences', indicating that strategies involving reinterpretation of the meaning of events were the major reappraisal strategy used (Fig. 1d). Negative affect ratings were substantially higher for negative than neutral pictures ('Look negative' – 'Look neutral'; $t = 63.8$, $P < 0.001$, Cohen's $d = 3.37$; Fig. 1e). Reappraisal substantially decreased negative affect compared to passive viewing ('Regulate negative' – 'Look negative'; $t = -17.5$, $P < 0.001$, $d = 0.92$). Eighty-one percent of individual participants showed regulation effects in the same direction as the group findings, indicating strong and robust effects.

Reaction times were also higher for 'Look negative' – 'Look neutral' (mean = 373 ms, $t = 22.2$, $P < 0.001$, BF > 10) and further increased for 'Regulate negative' – 'Look negative' (mean = 108 ms, $t = 6.42$, $P < 0.001$, BF > 10), suggesting that both affective processing load and reappraisal increase the demand on information integration during evaluation (Supplementary Fig. 2).

Brain systems identified by BF analysis

How separate are neural mechanisms of reappraisal from those involved in emotion generation? We tested this question by identifying 'Reappraisal only' brain regions activated during reappraisal ('Regulate' > 'Look negative'), with 10:1 odds in favor of an effect, BF ≥ 10 but not emotion generation ('Look negative' = 'Look neutral', with 10:1 odds in favor of the null, BF ≤ 0.1). We used the same BF thresholds in all subsequent tests and additionally required that voxels identified in each component map (for example, 'Reappraisal only') replicate all necessary effects in both studies independently (see 'Consensus map' subsection in Methods).

The consensus map for 'Reappraisal only' included 16 regions (Fig. 2a and Supplementary Table 1) located primarily in association cortex and cerebellum, including anterior lateral prefrontal cortex (adlPFC and avlPFC), temporal–parietal junction (TPJ), inferior parietal

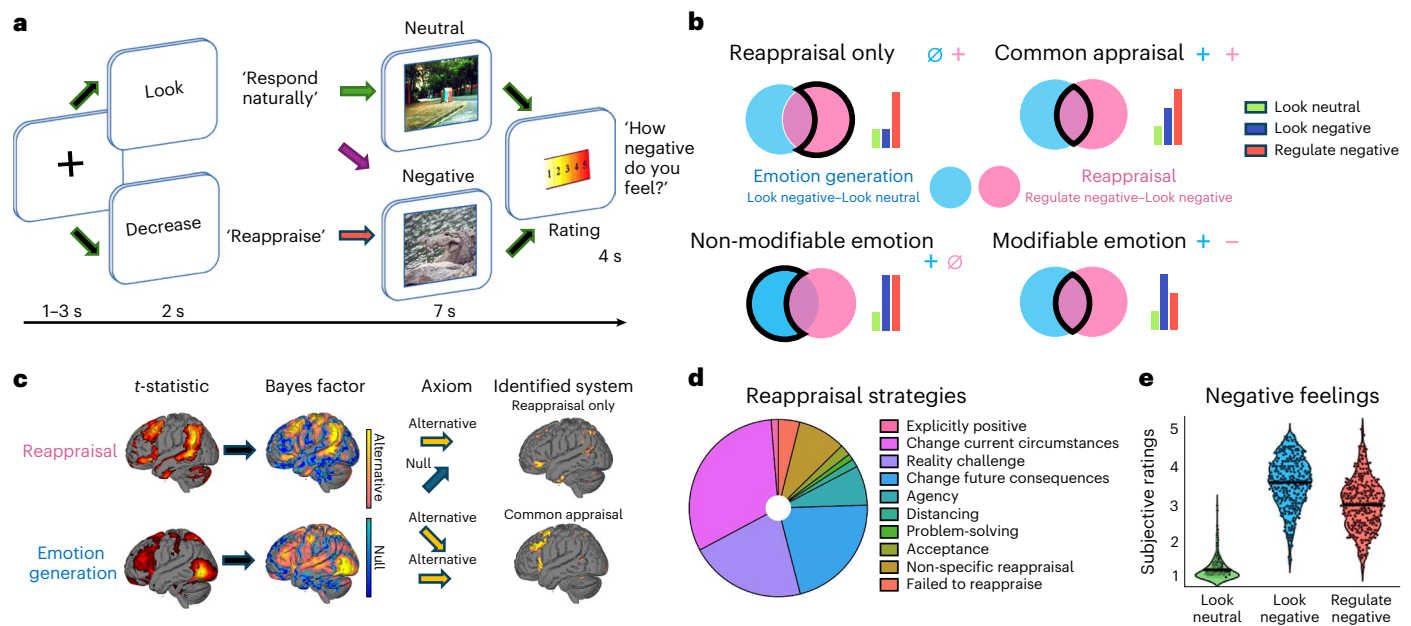


Fig. 1 | Overview of experimental paradigm and analysis pipeline. a, Trial structure. Sequentially, trials involved an appraisal cue ('Look' or 'Decrease', 2 s), image presentation (7 s) and negative affect rating (4 s). On a given trial, participants passively viewed a neutral image ('Look neutral'), passively viewed an aversive image ('Look negative') or reappraised an aversive image with the goal of downregulating negative affect ('Regulate negative'). **b**, Theoretical profile of activity across the three task conditions for each system component. Circles represent Venn diagrams of voxels responding to emotion generation ('Look negative' versus 'Look neutral', blue) and reappraisal ('Regulate negative' versus 'Look negative', pink). '+', '−' and '∅' denote relative activation (+) or deactivation (−), and '∅' denotes a null effect. The area of intersection represents an effect in both contrasts, and the non-overlapping areas are defined in part by null effects for one contrast. The portion of the Venn diagrams matching the axioms for

each component is outlined in black. **c**, Illustration of how BFs are combined across contrasts to identify voxels that conform to the axioms defining each system component, as exemplified by one dataset. *t*-statistics for group random effects contrasts are transformed to Bayes factor (BF) maps²⁸ and thresholded, with positive findings (BF > 10) indicated in yellow and null findings (BF < 1/10) indicated in blue. This threshold denotes strong evidence for alternative or null hypotheses¹⁰⁷. Conjunctions across thresholded BF maps identified voxels conforming to the axioms defining each system component. **d**, Specific reappraisal strategies. The percentage of particular reappraisal strategies used by participants. **e**, Average subjective negative ratings in each condition. Black dots represent the negative rating for individual participants averaged across trials in each condition.

lobe (IPL), posterior middle and anterior inferior temporal gyri (pMTG and aITG) and superior cerebellum. Among seven resting-state brain networks defined by Yeo et al.³², most voxels fell within the 'default mode' (47%), 'fronto-parietal' (31%) and 'ventral attention' (16%) networks. Although nearly all regions were replicated across both studies, a cluster in the SMA was uniquely identified in Study 1, and a cluster in the precuneus was uniquely identified in Study 2 (Supplementary Fig. 3).

We compared 'Reappraisal only' regions with those that fulfill the axioms for a 'Common appraisal' process, which are activated in the 'Look negative' – 'Look neutral' contrast and further increase during reappraisal ('Regulate negative' – 'Look negative'). 'Common appraisal' regions, shown in Fig. 2b, identified 14 regions, which is the largest set of brain regions of any component identified. These included dorsal and ventral lateral prefrontal cortices (dlPFC and vlPFC), superior parietal lobe (SPL), posterior superior temporal sulcus (pSTS), anterior insula (AI) and cerebellum. In contrast to 'Reappraisal only' regions, 'Common appraisal' regions included midline brain structures: dorsal medial prefrontal cortex (dmPFC), anterior midcingulate cortex (aMCC), SMA and precuneus. Most voxels were located in the frontal parietal network (51%), default mode network (32%) and dorsal attention network (10%). Nucleus accumbens (NAc) was uniquely identified in Study 1 but not in Study 2 (Supplementary Fig. 3).

A closer spatial comparison of 'Reappraisal only' and 'Common appraisal' regions is shown in Fig. 2c. In lateral cortical areas, areas for 'Reappraisal only' were located anterior and lateral areas proximal to 'Common appraisal' regions, suggesting a cortical gradient across areas sensitive specifically to goal-directed appraisal and those spontaneously engaged by affective stimuli.

Despite prior evidence suggesting that cue-induced anticipatory activity can influence reappraisal success (for example, ref. 33, but compare to ref. 34 for conflicting findings) and the value of proactive control more broadly³⁵, we found no reappraisal effects during the cue period in either 'Reappraisal only' ($t = 0.83$, $P = 0.41$, BF = 0.084) or 'Common appraisal' ($t = -0.76$, $P = 0.45$, BF = 0.080) regions, with BFs less than 0.1 providing strong support for null findings (Supplementary Fig. 4).

A second motivating question was to understand which emotion generation-related regions are downregulated during reappraisal. We divided regions activated during 'Look negative' – 'Look neutral' into 'Modifiable emotion' regions reduced by reappraisal ('Regulate negative' < 'Look negative') and 'Non-modifiable emotion' regions unaffected by reappraisal ('Regulate negative' = 'Look negative'). Most regions were non-modifiable, including all subcortical emotion-related regions (Fig. 3a). Non-modifiable subcortical emotion-related regions included the amygdala, PAG, parabrachial nucleus, ventral putamen and hypothalamus and ventral striatum and pallidum. Cortical regions included lateral orbitofrontal cortex (OFC), medial prefrontal cortex (mPFC) (gyrus rectus and midcingulate) and retrosplenial cortex, frontal area 6, inferior and superior parietal cortices, precuneus and ventral fusiform cortex extending into cerebellum (Supplementary Table 1). Most cortical voxels were located in the dorsal attention (37%), visual (16%) and limbic (16%) networks.

'Modifiable emotion generation regions' were located exclusively in the cortex and mainly in occipito-temporal and parietal regions (Fig. 3b). These included bilateral middle/inferior occipital cortex (including the lateral occipital complex (LOC)), postcentral gyrus

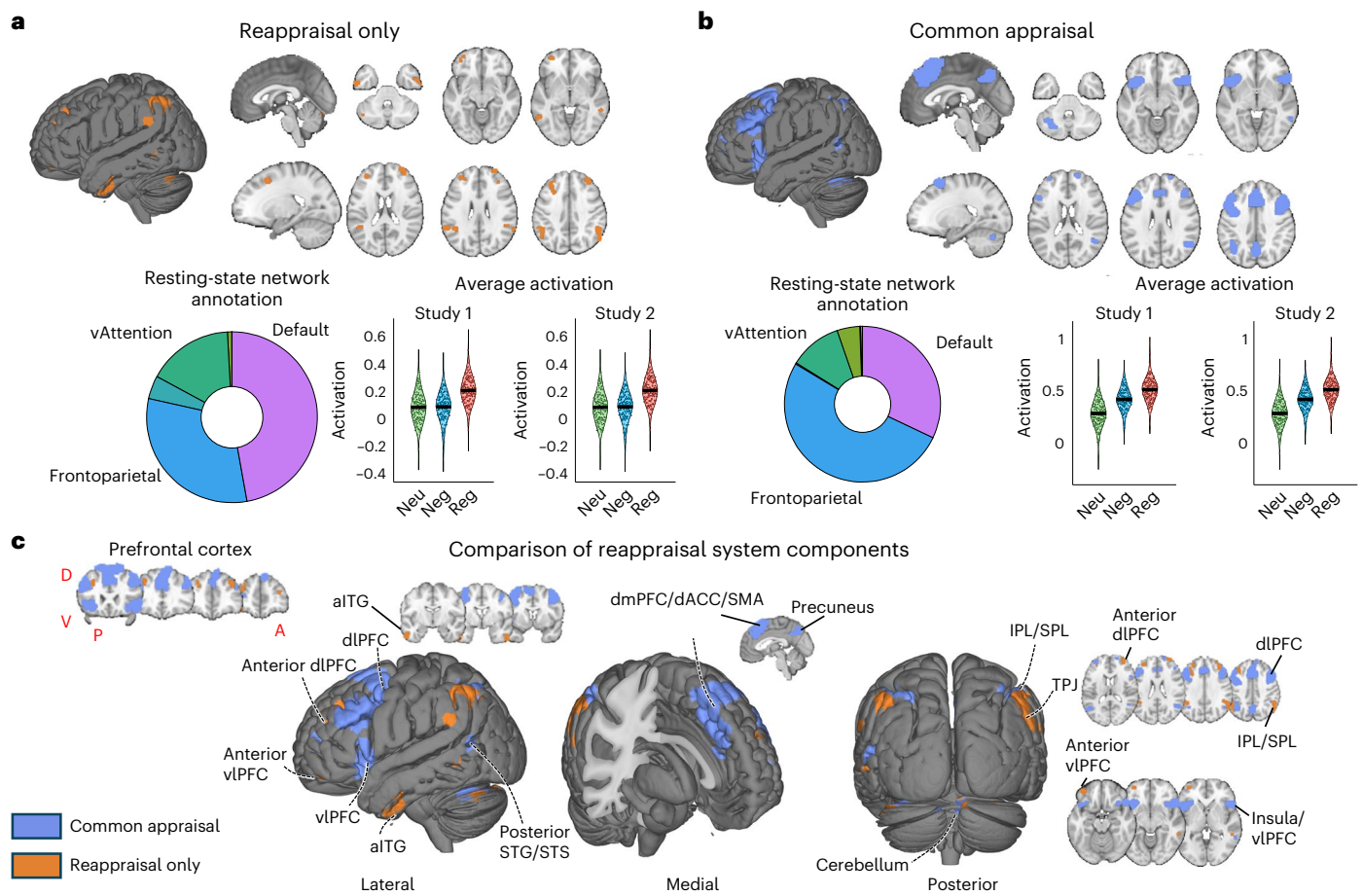


Fig. 2 | Consensus brain regions for reappraisal system components across both independent studies (see Methods for details on consensus maps).

a, Brain regions identified in ‘Reappraisal only’: areas involved in emotion regulation but not generating negative emotion. Pie charts show the percentage of cortical voxels from corresponding maps located in each of the seven resting-state cortical networks from ref. 32. Violin plots show the distribution of average from voxels identified in each consensus map. To visualize the within-subject condition-level effect, the average activation for each subject was substituted

with the grand mean, calculated across all participants. This method ensures the exclusion of between-participant variability and maintains within-subject variations across different conditions¹⁰⁸. Neu, Look neutral; Neg, Look negative; Reg, Regulate negative. **b**, ‘Common appraisal’ regions: areas positively activated in emotion generation and further increased by reappraisal. vAttention, ventral attention network. **c**, A direct comparison of ‘Reappraisal only’ (orange) and ‘Common appraisal’ (blue) regions. STG/STS, superior temporal gyrus/sulcus. D, dorsal; V, ventral; P, posterior; A, anterior; dACC, dorsal anterior cingulate cortex.

(including primary somatosensory cortex), intraparietal cortex and inferior frontal junction in posterior prefrontal cortex. Voxels were mostly located in the visual (51%), dorsal attention (31%) and somatomotor (12%) networks. Ventral mPFC and medial OFC regions were identified as modifiable only in Study 1, and a more dorsal mPFC region was modifiable only in Study 2 (Supplementary Fig. 3). No modifiable regions were identified in the subcortex.

A comparison between ‘Modifiable’ and ‘Non-modifiable’ regions (Fig. 3c) showed that ‘Modifiable’ regions were found mainly in retinotopic visual areas, such as the dorsal visuospatial cortex (V7) and intraparietal sulcus (IPS), ventral visual stream (V3, V4, ventral occipital (VO) and parahippocampal cortex (PHC)) and lateral visual cortex (lateral occipital (LO), medial superior temporal (MST) and middle temporal (MT)), whereas ‘Non-modifiable’ occipital regions were located mainly in non-retinotopic areas. Emotion generation regions in the superior parietal lobe were non-modifiable, whereas the precuneus and posterior IPL were modifiable. We also observed ‘Non-modifiable emotion’ regions in frontal regions, indicating the presence of emotion generation-specific structures in the frontal cortex.

We tested specifically for reappraisal effects in a priori amygdala subregions and PAG (Fig. 4). All amygdala subregions and the PAG responded robustly to negative images (‘Look negative’ – ‘Look neutral’, all $P < 0.0001$), indicating sensitivity to negative images, but no region

was significantly downregulated by reappraisal (‘Regulate negative’ – ‘Look negative’, all $BF < 0.1$). Effects in the basolateral and superficial amygdala were numerically in the opposite direction as predicted by the downregulation hypothesis. Separate models of early (0–2 s) and late (2–7 s) responses yielded similar findings (Supplementary Table 2). Additionally, we tested whether normative picture valence could influence amygdala regulation. Using a multi-level mixed-effect model, we found a null effect of ‘Regulation’ in all amygdala regions and a small ‘Valence’ × ‘Regulation’ interaction in centromedial (CM) and superficial (SF) amygdala ($d = 0.21$ and 0.27 , respectively, $P < 0.05$), indicating that the ‘Valence’ effect on amygdala activation is slightly attenuated (less negative) during reappraisal. The interaction in basolateral (BL) amygdala was not significant ($d = 0.18$ and $P = 0.085$). However, the BFs for these effects do not support a strong conclusion ($BF < 1/3$ in BL and close to 1 in CM and SF). Coupled with the null effects of ‘Regulation’, this suggests that amygdala activation could be marginally downregulated by reappraisal for the most aversive images but upregulated under reappraisal for the least aversive images—but if such effects exist, they are very small (Supplementary Fig. 5).

Correlation with reappraisal success

Individual differences in reappraisal success were correlated with reappraisal-related activity in the identified system components

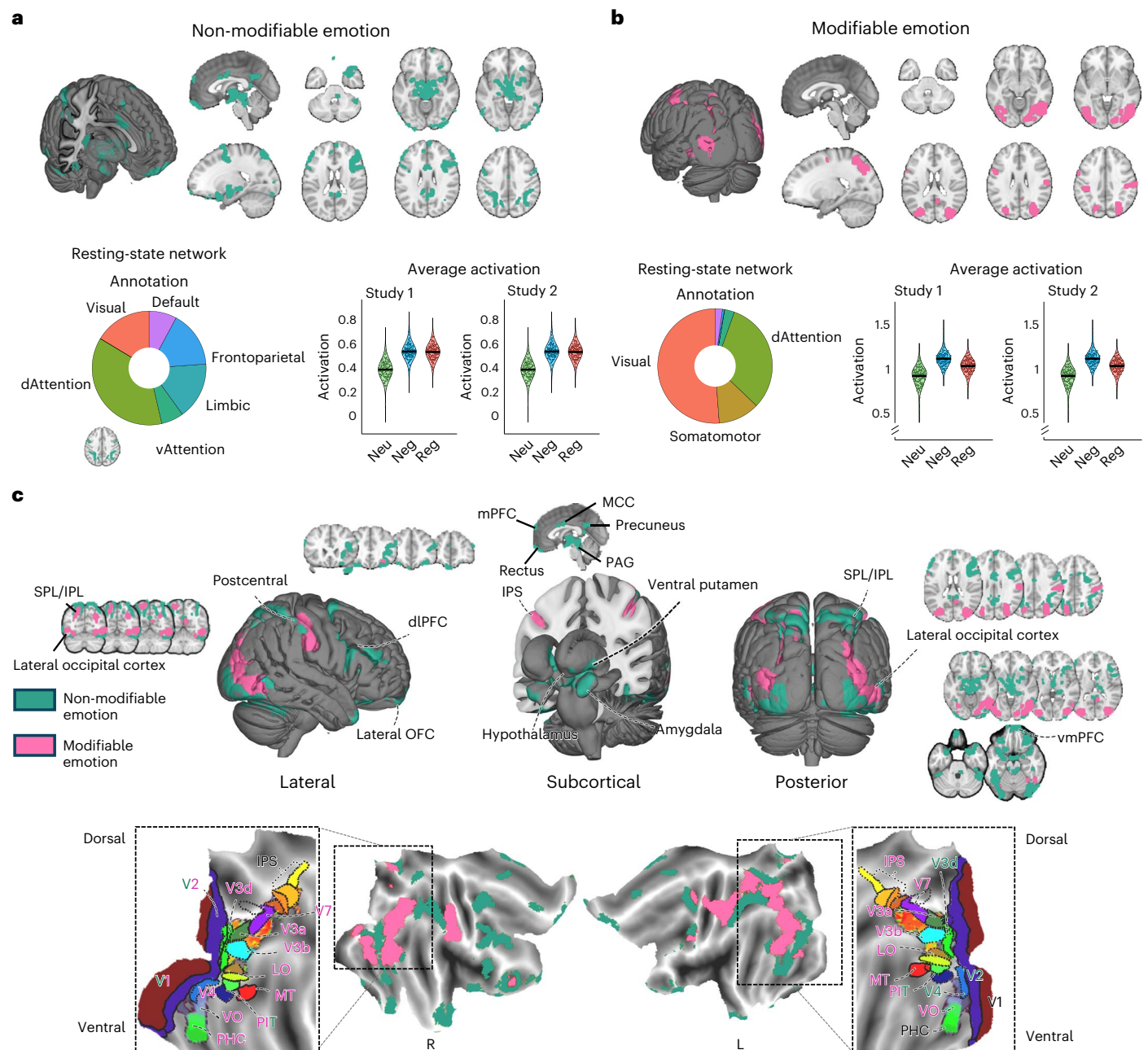


Fig. 3 | Consensus brain regions for emotion generation-related system components across both independent studies. Brain maps, pie charts and violin plots are as in Fig. 2. **a**, Non-modifiable emotion generation brain regions: areas involved in emotion generation but not downregulated by reappraisal. Neu, Look neutral; Neg, Look negative; Reg, Regulate negative. **b**, Modifiable emotion generation brain regions: areas involved in emotion generation and downregulated by reappraisal. dAttention, dorsal attention network. **c**, A direct

comparison between modifiable (pink) and non-modifiable (teal) emotion-related regions. The bottom panels show regions on flattened cortical maps, with visual cortical areas labeled for reference¹⁰⁹. Modifiable and non-modifiable retinotopic visual regions¹¹⁰ are shown in pink and teal fonts, respectively, and other areas are shown in black font. IPS, intraparietal sulcus; MCC, midcingulate cortex; PIT, posterior inferior temporal; V1, primary visual cortex; V2, V3 and V4, intermediate visual cortex.

(Fig. 5a), including increased activity in ‘Reappraisal only’ ($r = 0.2$, $P = 0.0002$, adjusted $R^2 = 0.039$) and ‘Common appraisal’ ($r = 0.14$, $P = 0.008$, adjusted $R^2 = 0.02$) systems and decreased activity in the ‘Modifiable emotion generation’ system ($r = -0.14$, $P = 0.008$, adjusted $R^2 = 0.017$). Correlations in the ‘Non-modifiable emotion generation’ system were not significant ($r = 0.015$, $P = 0.77$, adjusted $R^2 = -0.002$). These results remained consistent after adjusting for sex and age (Supplementary Fig. 6). Prediction of reappraisal success using best-subsets regression (Mallows’s C_p (ref. 36)) across the four components yielded a model including ‘Modifiable emotion generation’ and ‘Reappraisal only’ system components ($r = 0.26$, adjusted $R^2 = 0.063$, $P = 0.000003$,

Akaike information criterion = 697; Supplementary Table 4). Thus, activation of reappraisal-selective ‘generator’ regions and reductions in modifiable cortical ‘target’ regions both explain unique variance in reappraisal success.

To complement our systems-focused model, we examined correlates of reappraisal success across the whole brain. Previous brain correlates of reappraisal success were primarily tested in samples of $n < 70$ (refs. 37,38), but correlations did not stabilize to a 95% confidence interval of ± 0.1 until $n > 161$ (ref. 39). We sought to revisit this question with a larger sample size ($n = 358$ across both studies). As shown in Fig. 5b, reappraisal success was correlated with increased

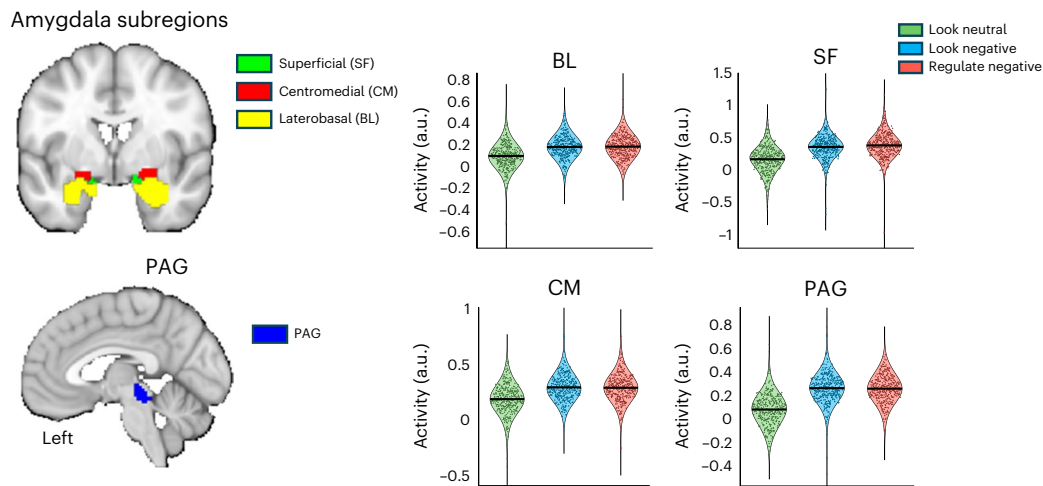


Fig. 4 | Region of interest analysis on amygdala and PAG. Amygdala regions are defined based on ref. 111, and the PAG region was defined anatomically by T.D.W. in 2018. All regions are available in the Cognitive and Affective Neuroscience

Lab 2018 combined atlas (see <https://canlab.github.io/>). The brain activation distributions for each condition are shown in the violin plot. It is the same within-subject display as Figs. 2 and 3. a.u., arbitrary units.

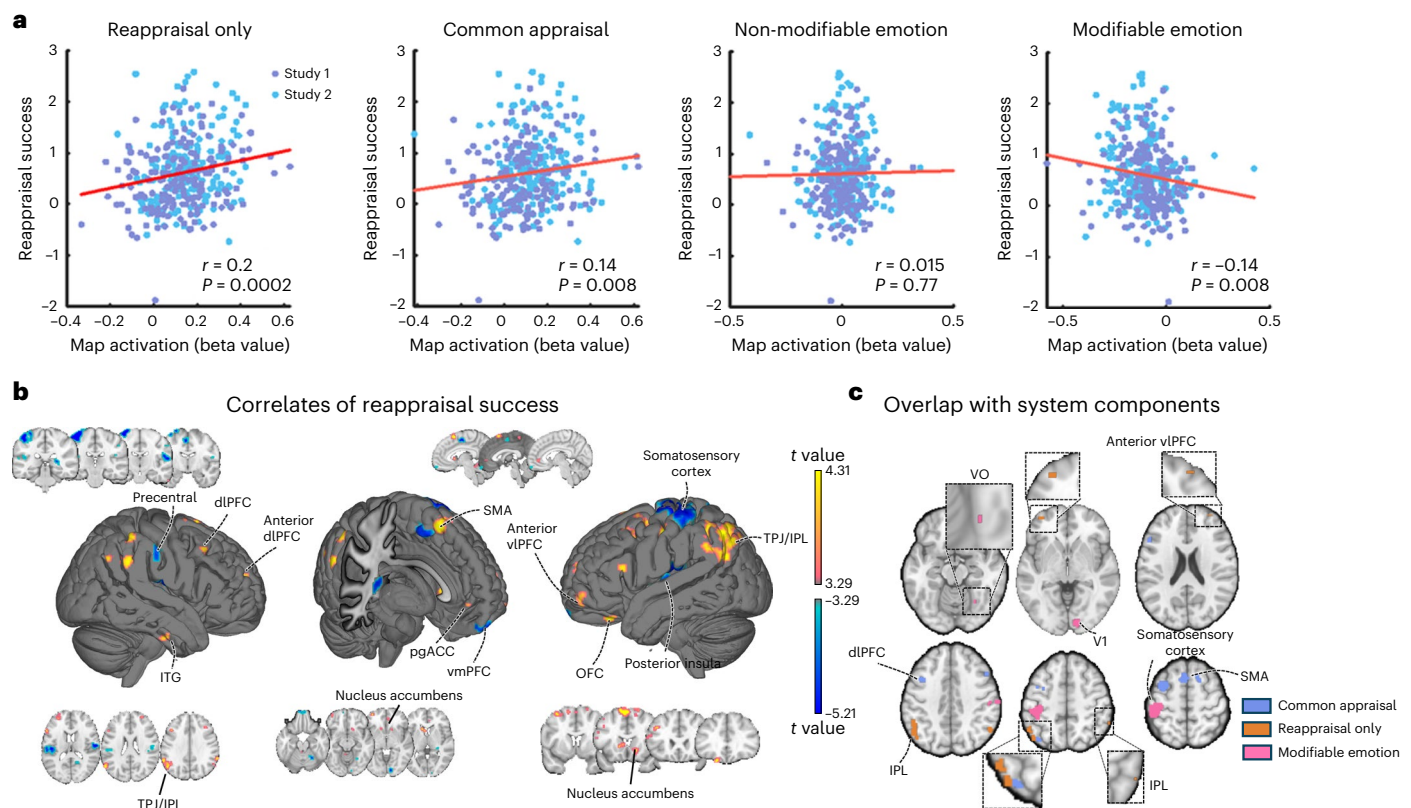


Fig. 5 | Correlation of reappraisal success with average activation from each component system. The brain activation was computed by the contrast between ‘Regulate negative’ – ‘Look negative’. **a**, Scatterplots depict the cross-participant correlation between reappraisal success and average activation within each system component. The correlation analyses are two-sided and computed using Pearson’s correlation method. Reappraisal success was correlated with increases in the average activation of ‘Reappraisal only’ and ‘Common appraisal’ systems and decreases in the average activation of the ‘Modifiable emotion generation’ system during reappraisal. These significant correlations remained significant after correction for multiple comparisons using Bonferroni correction (less

than 0.05/4). Best multiple regression model to predict reappraisal success: ‘Reappraisal only’ + ‘Modifiable emotion generation’ (supplementary result). Regions with positive correlation show similar spatial locations with ‘Reappraisal only’ regions. **b**, Whole-brain analysis of robust regression between reappraisal success and brain activation. FDR corrected, $q < 0.05$. Robust regression can minimize the influence of outliers. **c**, Illustrates the overlap regions between **b** and ‘Reappraisal only’, ‘Common appraisal’ and ‘Modifiable emotion generation’ regions. The first two system components overlap with regions showing positive correlations, whereas the third overlaps with regions showing negative correlations.

activation during ‘Regulate negative’ – ‘Look negative’ in multiple areas within lateral prefrontal cortex (anterior prefrontal cortex, dIPFC and vIPFC), SMA, ventromedial prefrontal cortex (vmPFC), TPJ and inferior

temporal gyrus, overlapping largely with ‘Reappraisal only’ (for example, anterior prefrontal cortex and TPJ) and ‘Common appraisal’ (for example, SMA, dIPFC and vIPFC) system components (Fig. 5c).

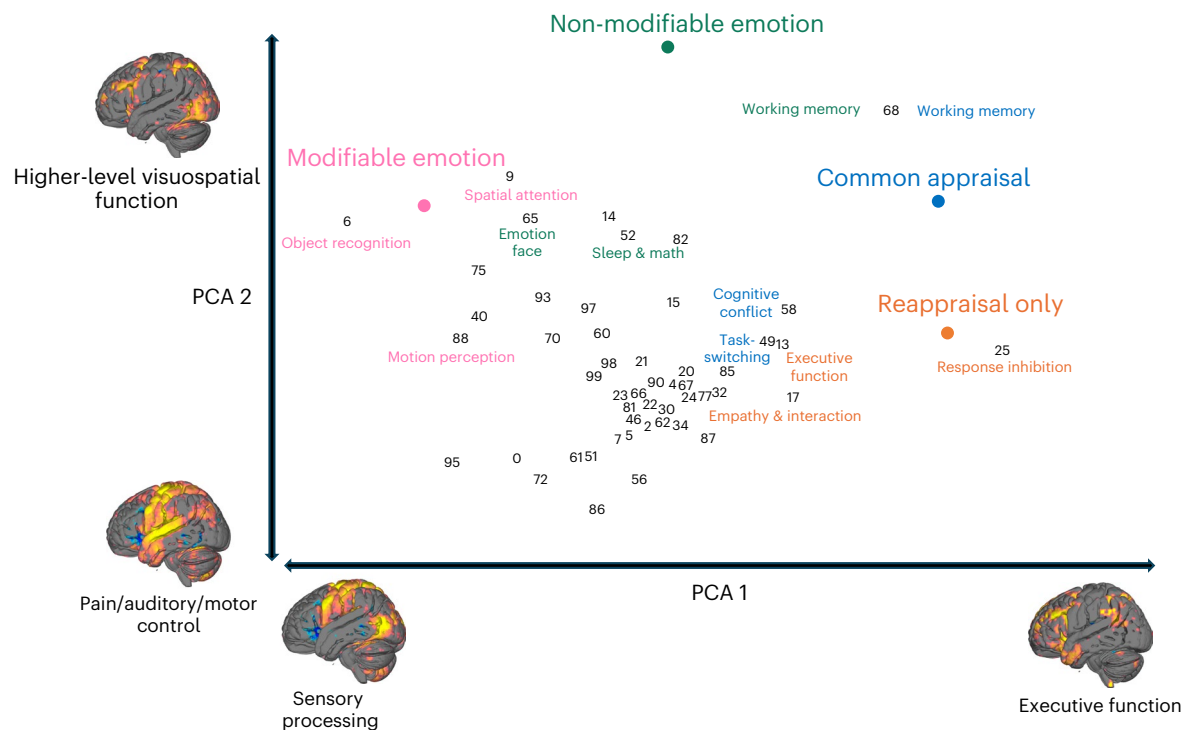


Fig. 6 | Function annotation using Neurosynth topic maps. Spatial similarities between Neurosynth topic maps and system components were assessed by Pearson correlation. This matrix was further normalized across topics within each system component map using Z-score normalization and decomposed into two dimensions using principal component analysis (PCA). All topic maps and system components are displayed on this two-dimension space given their loading on these two principal components. The brain map representing the two

ends of the principal components was generated by calculating the weighted average of the top five loaded (negative or positive) topic maps. The top three correlated topics for each system component are also displayed on the graph. Additionally, we computed the most selective topic for each system component (see Methods for details), which are as follows: ‘response inhibition’ for ‘Reappraisal only’; ‘cognitive conflict’ for ‘Common appraisal’; ‘emotion face’ for ‘Non-modifiable emotion’; and ‘object recognition’ for ‘Modifiable emotion’.

Most prefrontal regions that predict reappraisal success in ‘Reappraisal only’ are located in anterior vlPFC, whereas regions in ‘Common appraisal’ are located in more posterior vlPFC/operculum and dmPFC (Supplementary Fig. 7). Correlations also included increases in NAC, pregenual anterior cingulate cortex (pgACC) and superior precuneus.

Conversely, reappraisal success was associated with reduced activity in several regions overlapping with the ‘Modifiable emotion generation’ component, including the early visual areas V1 and VO and the sensorimotor cortex. Other regions include S2, posterior insula and operculum. These areas were not identified as emotion related or reappraisal related because they decreased with negative emotion (Supplementary Fig. 8). Activation of these areas has been associated with pain and other somatic processes and emotional feelings, including social rejection⁴⁰.

Overall, these findings validate smaller-sample reports of correlations in lateral PFC, dorsal mPFC/SMA and NAC^{37,41} and extend them by providing increased spatial precision and generalizability to a population-based sample^{42,43}.

Beyond the ratings themselves, we found that differences in rating reaction times correlated with brain activity contrasts in the ‘Common appraisal’ component for both emotion generation (‘Look negative’ – ‘Look neutral’, $r = 0.186$, $P = 0.004$) and regulation (‘Regulate negative’ – ‘Look negative’, $r = 0.14$, $P = 0.009$). However, these relationships were not significant in the ‘Reappraisal only’ regions ($r = 0.061$ and $P = 0.25$ for emotion generation; $r = 0.07$ and $P = 0.18$ for regulation). Correlations between ‘Common appraisal’ regions and reaction times imply a heightened information integration demand during both emotion generation and reappraisal. Additionally, these data offer inconclusive evidence on the relationships between brain components and the Emotion Regulation Questionnaire (Supplementary Fig. 9)

and individual differences in negative emotion ratings (‘Look negative’ – ‘Look neutral’; Supplementary Fig. 10).

Functional annotation of reappraisal system components

To test the relationships between component maps and other mental functions, cognitive control, we assessed the association between our component maps and 50 psychological topic maps related to mental processes—for example, face processing, inhibition and reward—from <https://neurosynth.org/> (ref. 30) (Supplementary Table 5). The ‘Reappraisal only’ system was most strongly and selectively correlated with the topics ‘response inhibition’, ‘executive function’ and ‘empathy & interaction’, suggesting a close relationship between reappraisal and executive control of responses and task sets (Fig. 6). The ‘Common appraisal’ system was most correlated with ‘working memory’, ‘task-switching’ and ‘cognitive conflict’ and unique to ‘cognitive conflict’, suggesting relationships with both executive function and value-related processes. ‘Non-modifiable emotion’ regions were most correlated with ‘emotional face’, ‘sleep & math’ and ‘working memory’ and unique to ‘sleep & math’. ‘Modifiable emotion’ regions were most strongly correlated with ‘object recognition’, ‘motion perception’ and ‘spatial attention’ and unique to ‘object recognition’. The latter findings suggest an overlap between the brain regions subserving affect and cognition^{44,45} and in particular between modification of emotion and selection of object features and properties in images. As with all such maps, the findings are hypothesis generating and likely reflect biases in the literature on structure–function associations.

Neurochemical annotation of reappraisal system components

Recent data-sharing efforts allow us to relate fMRI patterns to neurotransmitter receptor and transporter binding maps obtained in PET

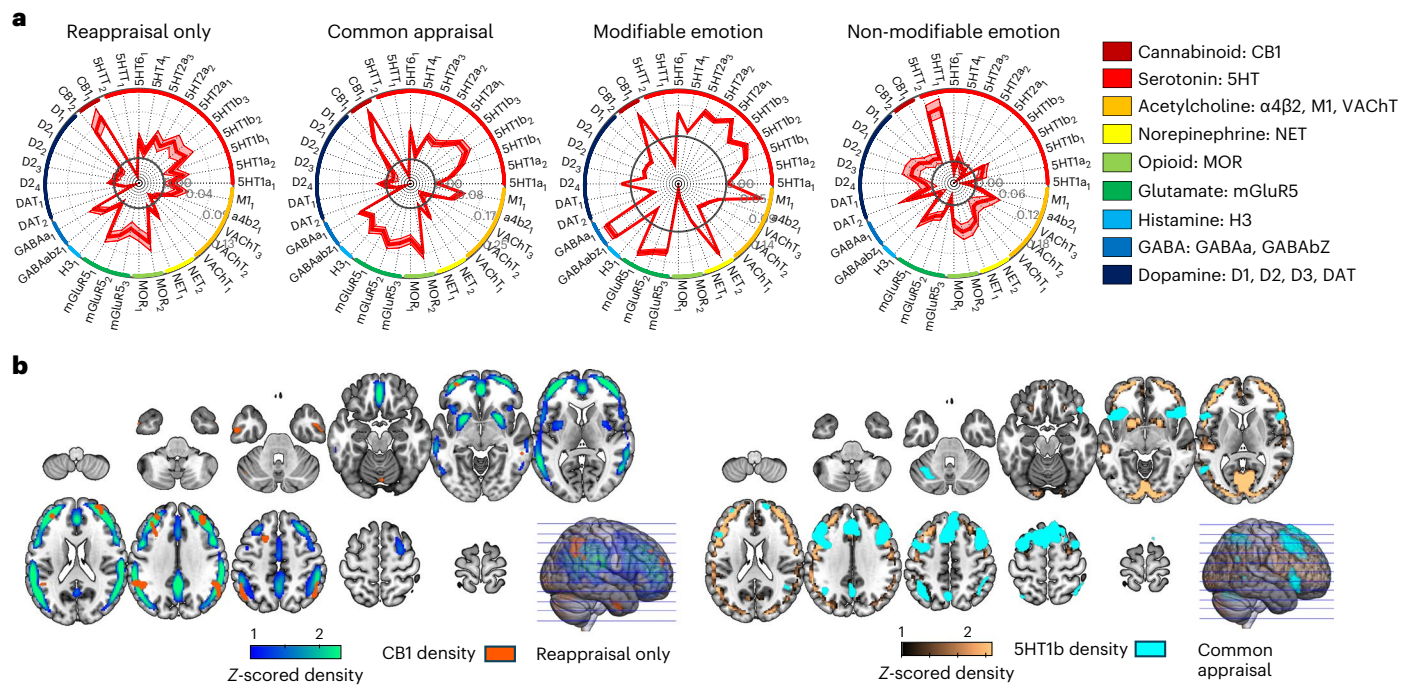


Fig. 7 | Neurochemical annotation of system component maps. a, Assessment of spatial similarities between individual neurochemical binding maps and system components using Pearson correlation. Error variability in component maps was evaluated through bootstrapping the entire BF and component generation analysis, yielding standard error bars for associations with neurochemical binding maps. A neurotransmitter system was considered significantly correlated only if the 95% confidence intervals of correlation coefficients from all replication studies were above zero. Neurotransmitter systems: cannabinoids (CB1 receptor), serotonin (5HT receptors 1a, 1b, 2a,

4 and 6 and transporter (5HTT)), acetylcholine (muscarinic M1, nicotinic a4b2 and transporter (VACHT)), norepinephrine (transporter (NET)), opioids (mu-opioid receptor), glutamate (metabotropic receptor mGluR5), histamine (H3), GABA (GABA-A and GABA-bZ receptor) and dopamine (receptors D1, D2 and transporter (DAT)). **b**, Illustration of two exemplary cases of spatial overlap between neurochemical binding maps and system components. Note that the neurochemical binding maps are normalized by Z-score normalization. The value on the color map merely reflects the relative density across voxels.

studies³¹. In a hypothesis-generating exploratory analysis, we correlated each system component map with baseline binding maps from 36 studies of eight different transmitter systems.

'Reappraisal only' and 'Common appraisal' patterns were both significantly associated with cannabinoid (CB1), serotonin (5HT2a and 5HT6), mu-opioids, glutamate mGluR5 and GABA-A, suggesting the involvement of similar overall neurochemical systems (Fig. 7 and Supplementary Table 6). 5HT1a was associated with 'Reappraisal only' but less consistently, and histamine H3 binding was correlated with 'Common appraisal' but not 'Reappraisal only', although only one histamine study was available, so replicability across PET maps could not be assessed. No other transmitter system was positively associated across PET studies.

'Modifiable emotion' regions were also associated with 5HT1b, 5HT2a, glutamate mGluR5 and GABA-A. Consistent negative correlations were found for mu-opioids, dopamine D1/D2 and VACHT, suggesting that the regional distribution of these neuromodulators is less likely to explain influences of reappraisal on activity. Inconsistent relationships across PET studies were found for CB1 and NET. In contrast, 'Non-modifiable emotion' regions exhibited a quite different transmitter profile, with consistent positive associations with 5HTT, NET, mu-opioids and dopamine D2. Thus, subcortical structures rich in dopamine, opioids and norepinephrine were not modifiable by reappraisal.

Discussion

Reappraisal is a foundational skill for regulating thoughts and behavior. However, to date, we have not had clear ways of separating brain activity related to reappraisal from activity related to emotion generation. As a result, it remains unclear whether reappraisal and affect generation are

similar or distinct neural processes, and we have not had brain measures to assess when and how strongly reappraisal-related neural processes are engaged. In this study, we took a step toward addressing these questions. We identified four theoretical reappraisal-related processes and found support for the existence of brain systems supporting each: (1) a reappraisal-specific process not engaged during emotion generation; (2) a common appraisal process engaged by both emotion generation and reappraisal; (3) negative emotion-related areas unaffected by reappraisal; and (4) negative emotion-related areas downregulated by reappraisal. The use of larger samples ($n > 350$) made it feasible to identify brain regions related to these processes based on evidence for both activation differences and equivalencies (using BFs) with adequate power and generalizability across two independent cohorts.

We found that reappraisal engages distinct regions not activated during emotion generation alone. These were located anterior to 'Common appraisal' regions in the prefrontal cortex, including adIPFC and avIPFC, and lateral and ventral to 'Common appraisal' regions in the parietal cortex, temporal cortex and cerebellum (including TPJ, aITG and cerebellar Crus I) and may play a unique role in developing human self-regulatory capacities. TPJ, aITG and adIPFC are among those with the largest expansion in humans versus other primates and during early-life human development⁴⁶. TPJ, aITG and anterior PFC also appear to be among the most integrative, trans-modal areas in the brain⁴⁷, positioned above unimodal sensory hierarchies and, thus, in a position to regulate them based on conceptual context.

Anterior dIPFC also has particularly high expression of human-accelerated genes⁴⁸ and is related to meta-cognitive monitoring⁴⁹, representation of contextual schemas⁵⁰, future-oriented thought⁵¹ and goal maintenance and planning⁵². Studies of effective connectivity⁵³ and lesion studies⁵⁴ also place it at the top level of cortical control

hierarchies. TPJ and anterior temporal cortex are interconnected with adIPFC⁵⁵ and may support similar functions. TPJ is a convergence zone for multiple cognitive processes, including attention and memory retrieval, and high-level conceptual processes, including perceived agency and abstract relations^{56,57}.

Although we identified reappraisal-selective regions, most regions activated during reappraisal were ‘Common appraisal’ regions also activated during negative emotion generation. These are presumably involved in emotion-generating appraisals and may be involved in spontaneous self-regulation in the absence of explicit regulatory instructions⁵⁸. These regions included medial prefrontal (dmPFC, aMCC, SMA and precuneus), lateral fronto-parietal (dlPFC, vlPFC, SPL, IPL and STS) and anterior insular cortices. These regions have been identified in previous meta-analyses of both emotion generation^{59,60} and reappraisal^{11,61} and are co-activated in studies of emotion⁶². These regions are located higher in cortical hierarchies⁶³ than sensorimotor regions but below reappraisal-selective regions.

Among ‘Common appraisal’ regions, dlPFC and vlPFC are most strongly associated with self-regulation¹⁰, and vlPFC correlates with upregulation of NAc and reappraisal success³⁷ (a finding that we replicate here; see discussion below). Although sometimes considered ‘cold’ regions involved in context representation and information selection⁶⁴, they may play a direct role in affective appraisal and regulation. For example, dlPFC directly correlates with trial-by-trial ratings of food craving during a self-regulation task⁶⁵, and vlPFC correlates with trial-by-trial ratings of negative affect and social rejection⁶⁶. Other ‘Common appraisal’ regions are also thought to integrate somatic, emotional and cognitive information to enable subjective affective awareness, particularly the anterior insula¹⁸, dmPFC⁶⁷ and aMCC⁶⁸.

In the context of generalizability, ‘Common appraisal’ and ‘Reappraisal only’ regions were correlated with individual differences in reappraisal success (‘Reappraisal only’ particularly so) and generalized across strategies (Supplementary Fig. 11). Several ‘Common appraisal’ regions—including preSMA, vlPFC, dlPFC and insula—are also engaged by other emotion regulation strategies, such as expressive suppression and distraction^{11,13}. ‘Reappraisal only’ and ‘Common appraisal’ regions were associated with Neurosynth meta-analytic topic maps for working memory and response inhibition, providing a link between appraisal and other forms of cognitive control. Associations with neurotransmitter maps³¹ show that prefrontal and parietal ‘Reappraisal only’ and ‘Common appraisal’ regions are among the cortical regions with the highest levels of serotonin receptors (including 5HT1b and 5HT2a) and cannabinoid (CB1) receptors. Serotonin systems are broadly implicated in mood, aggression, sleep and pain and have been associated with psychedelic drug effects (especially 5-HT2A)⁶⁹ and psychotherapy (5-HT1B⁷⁰ and 5-HT1A⁷¹). CB1 is also implicated in mood regulation⁷². These studies point to common substrates of appraisal-based and pharmacological interventions in brain neurochemistry.

These reappraisal-related regions provide new brain targets for interventions, including psychotherapy, self-regulation training, neurostimulation and neurofeedback^{73,74}. Using areas activated during reappraisal as targets may not be an effective strategy, because most areas activated by reappraisal are also activated by negative emotion generation and could increase negative affect as well as potentiate reappraisal. Instead, focusing on ‘Reappraisal only’ regions may be a better approach.

Another goal of emotion regulation research is to understand which affect-generating processes are targeted and downregulated by reappraisal and related cognitive interventions. In previous work, amygdala was reported as the primary target of emotion regulation^{10,75} and is frequently taken as an indicator of regulatory efficacy⁷⁶. However, the amygdala and other subcortical affect-related regions—particularly PAG, hypothalamus and ventral striatum and pallidum—were identified as ‘Non-modifiable emotion’ regions in the present study and were neither influenced by reappraisal nor correlated with reappraisal success.

This might indicate that subcortical affective processes are less modifiable by reappraisal than previously thought. They may be more rapidly engaged, occurring before cognitive intervention is applied⁷⁷, and be more automatic and less susceptible to conceptual information⁷⁸.

Previous studies are actually surprisingly mixed on whether they find amygdala downregulation. Although early studies and meta-analyses identified amygdala decreases^{10,75}, many recent studies^{79,80} and meta-analyses^{11,81} do not (see also ref. 82), and some report reappraisal-related amygdala increases^{79,83}. Importantly, neither individual studies nor coordinate-based imaging meta-analyses (CBMAs) formally consider (1) the base rate of null findings across studies and (2) analytic flexibility that can increase false-positive rates. For example, some studies finding amygdala downregulation do not specify amygdala regions of interest a priori, as we did in the present study, but rather rely on tests across hundreds of amygdala voxels at uncorrected thresholds. Unlike traditional meta-analyses, CBMAs do not formally test the average amygdala effect size across all studies, and meta-analytic findings are only indirectly influenced by the accumulation of null results. In addition, the use of lower thresholds in amygdala inflates the number of positive findings in amygdala relative to other regions and reduces the effects of null findings on the meta-analytic results. Thus, there is a need for highly powered confirmatory tests of precisely defined regions and analytic procedures⁸⁴, such as those provided here. With $n = 358$ participants, our study is highly powered to detect amygdala reductions at the voxel level, with 80% power to detect even small effects of $d = 0.2$ or greater at typical $q < 0.05$ FDR-corrected voxelwise thresholds ($P < 0.002$, two tailed⁸⁵).

In addition, amygdala downregulation may depend on the strategy used, goals to increase or decrease emotion and how effectively strategies are deployed. For example, Dorfel et al.¹³ found that detachment was superior to reappraisal in decreasing amygdala activity, and two meta-analyses suggest that psychological distancing produces more consistent amygdala decreases than reinterpretation⁸¹ or extinction⁸⁶. The dominant strategies that our participants adopted were reinterpretation-based, including altered appraisals of images’ reality (‘reality challenge’), current circumstances and future consequences and the agency of those depicted (Fig. 1d). We did not find relationships between amygdala decreases and either the strategy used or reappraisal success, although such relationships were significant for other regions, including decreases in the ‘Modifiable emotion’ cortical system and increases in ‘Reappraisal only’, ‘Common appraisal’ and other regions, including NAc (Fig. 5). However, we did not systematically investigate less common strategies, such as psychological distancing.

Other factors may influence the strength of amygdala downregulation, including the timing of regulation effects, previous exposure or practice reappraising particular stimuli and potential interactions with population characteristics, such as age and sex. In terms of timing, Pierce et al.⁸⁰ found reappraisal effects only during a late post-image-viewing period (see also ref. 87). In the present study, we did not find regulation effects on either early or late picture-viewing periods, but the study was not designed to investigate effects during the post-viewing rating period. Other studies have found that amygdala responses during passive viewing are reduced when participants previously reappraised a particular image^{88,89}. Thus, practice regulating specific stimuli may be important. Studies vary on whether they report how items are re-used, and future studies should report this explicitly. In terms of effects of stimulus valence, we did not find that amygdala downregulation depended appreciably on how aversive an image was, similar to previous results⁹⁰, although we did find weak evidence for stronger effects on the most aversive images. Our images were restricted to a narrow range of highly aversive images, and this effect should be studied across a wider range of stimulus valence in future studies.

Finally, amygdala downregulation may vary across samples and populations for other reasons as well. Our population in the

present study was a midlife community sample of adults from Western Pennsylvania, which was older than many previous reappraisal studies based largely on college student populations. We did not find any interactions between reappraisal effects and sex or age in the present study, but it is possible that young, university-based samples could show different effects.

The regions most strongly downregulated by reappraisal ('Modifiable emotion regions') were in the visual and somatosensory cortices. One explanation may be that participants shifted their attention from emotional to non-emotional content during reappraisal⁹¹, leading to decreased activation in visual cortices during reappraisal. However, previous research has demonstrated successful reappraisal even when visual attention is controlled^{92,93}. In addition, shifting attention from emotional to non-emotional content results in decreased activity in early visual cortex in previous studies⁹⁴, whereas 'Modifiable emotion' regions are localized to higher-order visual regions (for example, V3, V4, V7, MT, VO, MST and MT) and parietal cortex. These regions are related to visual properties, such as object and scene representation⁹⁵, but also encode semantic meaning and context⁹⁶. Therefore, reappraisal may impact semantic and conceptual representations in these areas rather than simply reflecting shifts in attended location.

Extending this view, visual and somatosensory streams are increasingly seen as integral to affect generation. Recent work suggests that human sensation and emotion are inextricably linked⁹⁷, and predicting perceptual and action demands is an essential part of embodied views of emotion⁹⁸. Previous studies found that emotion triggers distinct neural representations in the visual cortex^{99,100}. These neural representations are valence specific^{101,102}, and different visual cortical patterns encode at least five distinguishable emotion categories⁹⁹. Other types of affect appear to be embedded in early sensory pathways as well. For example, a recent study found a superior collicular-visual pathway specific to encoding visually generated negative affect, an inferior collicular-auditory pathway specific to aurally generated negative affect and separate somatosensory representations for thermal and mechanical pain²². Additionally, somatosensory and posterior insular areas downregulated by reappraisal may be related to interoception¹⁰³. A recent study on emotion regulation also found a similar downregulation¹⁰⁴.

Other 'Non-modifiable emotion' regions are distributed throughout the brain, including dlPFC, OFC, IPS, premotor cortex and early visual areas, implying that cognitive reappraisal affects only a subset of affect generation processes. For example, some parts of dlPFC (particularly the inferior frontal junction and the inferior frontal sulcus), dmPFC and OFC (Brodmann's area 11) are engaged during negative affect but not modifiable. Some dlPFC areas that overlap 'Common appraisal' are similar to those that predict trial-to-trial variation in affective value¹⁰⁵. OFC regions are also among those most closely associated with appetitive and aversive value²⁰, particularly in relation to contextual variables, and in value-based reinforcement learning¹⁰⁶. The relative 'immunity' of these regions to reappraisal suggests that cognitive reappraisal may not be entirely effective in regulating the evaluation of affective stimuli. In addition, these findings update the commonly held view that prefrontal cortices are 'cold' cognitive structures particularly influenced by self-regulation and provide evidence for a more nuanced pattern.

In conclusion, our findings indicate that emotion generation and reappraisal depend on many of the same cortical regions, in line with psychological theories that posit overlap between affective appraisal and reappraisal. However, reappraisal involves a distinct set of brain regions situated at the highest levels of cortical hierarchies and is associated with executive cognitive functions, particularly managing hierarchies of goals over long timescales. They are also enriched in serotonin (5HT-1B and 5HT-2A) and CB1 receptors, which may preferentially subserve and interact with reappraisal. The target systems downregulated by reappraisal are limited mainly to perceptual, semantic and interoceptive processes related to the construction of affective value.

We found no effects on subcortical drivers of affective experience (for example, amygdala, PAG and hypothalamus) or on lateral prefrontal and orbitofrontal areas related to value construction. These findings inform on the systems involved in reappraisal and those involved in altered affective experience and provide new targets for assessment and intervention.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-024-01605-7>.

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Methods

Participants and paradigm

Participants were derived from two study cohorts: the Adult Health and Behavior project, phase 2 (AHAB-2; Study 1), and the Pittsburgh Imaging Project (PIP; Study 2)⁴³. Both studies recruited participants from the Greater Pittsburgh area. In total, 182 participants (95 males and 87 females, mean age = 43.45 ± 7.26 years) from Study 1 and 176 participants (87 males and 89 females, mean age = 40.58 ± 6.33 years) from Study 2 are included in the current study. These participants completed an emotional picture viewing and reappraisal task. Details of the task were provided in a previous study¹¹². During the task, participants viewed 30 unpleasant and 15 neutral International Affective Picture System (IAPS) images¹¹³. First, a 2-s cue ('Look' or 'Decrease') instructed participants to either look and attend to images ('Look') or decrease their negative feelings about aversive images by reappraising their circumstances and content ('Decrease'). Following the cue, images were presented for 7 s. After picture presentation, participants had 4 s to rate their negative emotion on a five-point scale (1 = neutral, 5 = extremely unpleasant), followed by a randomized rest period (1–3 s). E-Prime software was used to present the task and collect behavioral ratings. The task comprised 15 'Look neutral' trials, 15 'Look negative' trials and 15 'Decrease negative' trials. Eleven unpleasant images for 'Decrease negative' trials, 13 unpleasant images for 'Look negative' trials and two neutral images overlapped across Study 1 and Study 2. Images were shown in a pseudo-random order. No more than two identical instructional cues ('Look' or 'Decrease') were presented consecutively. No more than four negative images were shown consecutively. Data collection and analysis were not performed blinded to the conditions of the experiments. In Study 1, mean normative arousal and valence values for pictures assigned to 'Look negative', 'Regulate negative' and 'Look neutral' conditions were (6.16 ± 0.52 , 2.13 ± 0.37), (6.21 ± 0.67 , 2.01 ± 0.35) and (2.65 ± 0.40 , 5.12 ± 0.57), respectively. Values in Study 2 were (6.14 ± 0.53 , 2.10 ± 0.37), (6.37 ± 0.52 , 2.02 ± 0.36) and (3.44 ± 0.37 , 5.97 ± 0.35), respectively. Aversive images were thus similar in normative valence, and substantially more aversive than neutral images.

One concern regarding this task paradigm is that participants might not attend to the pictures and look away during picture viewing period. Because we did not record eye tracking or related data to confirm fixation on the pictures for the entire viewing period, we applied two additional analyses to test the potential influence of eye movement. In the first analysis, we compared our main results with a Neurosynth term map that associates with 'eye movement' and found limited spatial correlation (Supplementary Fig. 12a). In the second analysis, we applied a recently developed deep neural network model for predicting eye movements based on fMRI data, DeepMREye¹¹⁴. We analyzed correlations between model-derived eye movement estimates and task contrasts of interest. Specifically, we first calculated a measure of absolute eye movement (change in gaze position) given DeepMREye position coordinates x and y : $m = \sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}$, where t indexes time. Second, we regressed m on the task regressors for each participant and calculated contrasts of interest. Third, we performed a second-level random effects analysis on the contrasts, assessing 'Regulate negative' versus 'Look negative' and 'Look negative' versus 'Look neutral' differences in absolute eye movement. We did not find strong evidence that eye movement is systematically different across task conditions (Supplementary Fig. 12b). These results suggest that eye movement does not play a substantial role in our findings, although the absence of eye movement data to validate and fine-tune the model for the second analysis is a limitation.

Reappraisal strategies

A post-task inventory was administered to participants to assess different emotion regulation strategies that may have been used during the fMRI task paradigm, as detailed previously¹¹⁵. As part of the inventory, participants were asked to describe the strategies that they used

to change their feelings in the 'Regulate negative' condition. Using an open-ended response format, participants were asked: 'What sorts of things did you tell yourself to try to help you feel differently in response to the negative pictures?' Open-ended responses were coded according to standardized procedures detailed in previous work⁵. Responses were coded as reappraisal if they fell into any of these categories: (1) 'explicitly positive', (2) 'change in current circumstances', (3) 'reality challenge', (4) 'change of future consequences', (5) 'ascribing agency to a person capable of changing the circumstances depicted in the scene', (6) 'distancing', (7) 'problem-solving', (8) 'acceptance' and (9) 'non-specific reappraisal'. A final category was 'failed to reappraise', which included a failure to use any form of reappraisal (for example, looking away or not thinking about the IAPS scene, using suppression or using a strategy that was not cognitive).

fMRI data acquisition and pre-processing

MRI imaging data from Study 1 and Study 2 were collected on the same 3T Trio TIM whole-body scanner (Siemens) using a 12-channel, phased-array head coil. fMRI acquisition parameters were as follows: FOV = 205×205 mm, matrix size = 64×64 , TR = 2,000 ms, TE = 28 ms and FA = 90° .

fMRI data were pre-processed with statistical parametric mapping software (SPM12; <http://www.fil.ion.ucl.ac.uk/spm>). Functional images were realigned to the first image of the series with a six-parameter rigid body transformation. Before realignment, slice timing correction was applied to the IAPS fMRI task data to account for acquisition time variation in this event-related design. Realigned images were co-registered to each participant's skull-stripped and biased-corrected MPRAGE T1 image. Co-registered images were then normalized to Montreal Neurological Institute (MNI) space and interpolated to $2 \times 2 \times 2$ -mm voxels. Normalized images were then smoothed by a 6-mm full-width-at-half-maximum (FWHM) Gaussian kernel.

First-level analysis

Within-subject task effects and contrasts were estimated with a general linear model (GLM). The task conditions ('Look neutral', 'Look negative' and 'Regulate negative') were separately modeled with boxcar functions and convolved with the canonical hemodynamic response function (HRF) using SPM12 software. Regressors included one for the picture-viewing period in each task condition and one for each of the cue and rating periods collapsed across task conditions. We chose this method to minimize multicollinearity between the picture-viewing regressors and to maximize stability of individual person-level estimates. To test the robustness of the method's findings, we also repeated the analyses with a second first-level model that included separate regressors for the 'Look' and 'Regulate' cues. The main results remained robust and very similar to those that we report here, with no major changes in the conclusions (see Supplementary Fig. 4 for cue period results). The timing and duration of stimulus epochs were modeled according to the onset and duration of the stimulus presentation for each event. A high-pass filter of 1/180 Hz was applied to remove low-frequency temporal drift. In addition, nuisance regressors included six head motion parameters (x , y , z , roll, pitch and yaw), spike indicator regressors for potential outliers (with a regressor for each individual potential outlier coded as 1 and other volumes coded as 0), and time series of white matter and cerebrospinal fluid. Time series of fMRI responses within canonical eroded white matter and cerebrospinal fluid masks were extracted using CANlab tools (<https://github.com/canlab/CanlabCore>). The six head motion covariates were extracted from realignment parameters. Images were defined as spikes if they were outside the 95% confidence regions of the cloud of images in multidimensional space (Mahalanobis distance). The number of excluded potential outliers per participant was 19 ± 7.7 for Study 2 and 17 ± 8.0 for Study 1, representing approximately 5% of the fMRI images.

Contrasts of interest included ‘Look negative’ – ‘Look neutral’ and ‘Regulate negative’ – ‘Look negative’. These two contrasts were further applied to each individual participant’s parameter estimate values for subsequent BF analysis. The traditional T contrast maps for these contrasts of interest can be found in Supplementary Fig. 13.

The system identification approach

We established an axiomatic method based on BF to identify the proposed system components. Each process in our theoretical model must satisfy axioms—properties that must hold for any signal that encodes the component—that are defined based on the conjunction of both activation effects and null effects across two contrasts: (‘Look negative’ – ‘Look neutral’) (‘Emotion generation’) and (‘Regulate negative’ – ‘Look negative’) (‘Reappraisal’). BFs are applied to quantify and compare the support evidence for both null and alternative hypotheses for each contrast. BF value reflects the likelihood ratio between the alternative and the null. BF values in the current study are computed using JZS prior, and the method was provided in a previous study²⁸. In the method, t -statistics and sample size are required to compute the BF value. The equation is provided in the following:

$$B_{01} = \frac{\left(1 + \frac{t^2}{v}\right)^{-(v+1)/2}}{\int_0^\infty (1 + Ngr^2)^{-\frac{1}{2}} \left(1 + \frac{t^2}{(1 + Ngr^2)v}\right)^{-(v+1)/2} (2\pi)^{-1/2} g^{-3/2} e^{-1/(2g)} dg}$$

where N stands for the sample size; t denotes the t -statistic; and v is the degrees of freedom ($N - 1$ for most analyses). r denotes the scale factor. Note that the scale factor used in current study is $r = 0.707$. This scale factor suggests that a moderate effect size is expected as a prior, as outlined by ref. 28. It is typically taken as the default setting for the computation of the BF. See Supplementary Fig. 14 for testing the effect of different scale factors on main results.

Following the tutorial in ref. 116, we used a strict threshold of $BF > 10$ for evidence favoring the alternative hypothesis and $BF < 1/10$ favoring the null hypothesis, which is considered strong evidence. Note that a BF of 10 corresponds to a P value threshold of $P = 0.0018$ in current datasets, which satisfies (and is substantially more rigorous than) the $q < 0.05$ FDR-corrected threshold for both ‘Regulate negative’ – ‘Look negative’ and ‘Look negative’ – ‘Look neutral’. Using these thresholds, we defined system components in both datasets in the following axioms:

‘Reappraise only’ voxels have $BF > 10$ and $t > 0$ for ‘Reappraisal’ and $BF < 1/10$ for ‘Emotion generation’. In addition, these voxels have positive activation in the ‘Regulate negative’ condition.

‘Common appraisal’ voxels have $BF > 10$ and $t > 0$ for both ‘Reappraisal’ and ‘Emotion generation’. In addition, these voxels have positive activation in the ‘Look negative’ condition.

‘Non-modifiable emotion generation’ voxels have $BF > 10$ and $t > 0$ for ‘Emotion generation’ and $BF < 1/10$ for ‘Reappraisal’. In addition, these voxels have positive activation in the ‘Look negative’ condition.

‘Modifiable emotion generation’ voxels have $BF > 10$ and $t > 0$ for ‘Emotion generation’ and $BF > 10$ for ‘Reappraisal’ with a negative effect ($t < 0$, reduced activity during reappraisal). In addition, these voxels have positive activation in the ‘Look negative’ condition.

Note that the thresholds are asymmetric to the null and alternative hypothesis. The null is bounded at a minimum where the means are identical ($t = 0$). The range for t value would be narrower ($BF < 0.1$) for the null compared to strong evidence in favor of the alternative ($BF > 10$). This range is dependent on sample size. A sample size of $n = 120$ is the minimum number to find a stable t range for the null hypothesis (Supplementary Fig. 1). A larger sample size is better to find robust evidence in favor of the null hypothesis.

Consensus maps

As shown in Supplementary Fig. 3, identified systems from Study 1 and Study 2 were located in similar regions but did not completely overlap. To identify voxels in similar locations across both studies, we first conducted spatial smoothing using a 3-mm Gaussian kernel on each system component map (binary map). Next, to test the extent of overlap between Study 1 and Study 2, we multiplied the voxel values in smoothed maps for Study 1 and Study 2 to yield a product map. Higher product map values indicate higher degree of spatial overlap on a certain voxel. A binary consensus map was constructed by setting a minimum threshold for the product map. In the present study, we used 0.01 as a threshold to keep as much spatial similarity and a similar number of identified voxels in the consensus map compared with the individual study maps. In both individual studies and the consensus map, we retained clusters ≥ 15 contiguous voxels.

To characterize the consensus maps, we tested if the voxels revealed in consensus maps overlap with or are close to the identified system from Study 1 and Study 2. Specifically, we tested the distance between voxels in the consensus map and their nearest voxels in each of the thresholded BF maps from Study 1 and Study 2. For each voxel, the maximum distance across Study 1 and Study 2 was recorded. We computed this maximum distance for all voxels in the consensus map. This distance was less than 4.5 mm for all consensus component maps, indicating a voxel in both studies within 4.5 mm of each consensus map voxel. Specifically, the maximum deviating distance for each system component was ‘Common appraisal’, 4.4 mm; ‘Reappraisal only’, 4 mm; ‘Non-modifiable emotion’, 4.4 mm; and ‘Modifiable emotion’, 4.4 mm.

Neurosynth dataset

To provide an unbiased and data-driven based method to annotate the identified system components, we used the Neurosynth database (<http://neurosynth.org>), which relies on reported activation pattern from 11,406 studies³⁰. The large scale of the Neurosynth dataset is able to compensate for the possible random error reported in a single study, providing robust assignment of brain function in a given region.

We selected 50 psychological topics (for example, face processing, inhibition and reward) from a set of 100 topics (V4-topics-100) based on 11,406 studies, excluding topics related to psychological content such as ‘stimulation’, ‘task’ or ‘paradigm’. The topic maps are computed using reverse inference, which denotes probability of the topic given activation. Detailed methods to generate these topic maps can be found in Poldrack et al.¹¹⁷. To calculate the spatial similarity between topic maps and system components, topic maps were resampled into the same space as system components. Spatial similarity was assessed via Pearson correlation between topic maps and identified systems in a pairwise fashion. Due to the large variation in voxel counts across system components (‘Reappraisal only’, ‘Common appraisal’, ‘Non-modifiable emotion generation’ and ‘Modifiable emotion generation’), the correlation coefficients varied across components. To eliminate systematic spatial extent-related differences, we normalized the correlation coefficients within each system using Z -score normalization to make sure that each system component obtains the same average correlation with 50 topics, which represents the relative correlation across all topics. A principal component analysis was conducted on the normalized correlation matrix, and the relative locations of topics and system components were plotted in the space of the first two principal components. The top three correlated topics for each system component are displayed.

We also reported the topics most selectively associated with one system component compared to the others’ components. Specifically, a selectivity score for each topic was computed as the difference between the best correlated system component and the second best one. For each system component, we identified the topic that had the highest selectivity.

PET tracer neurotransmitter receptor density dataset

We investigated the spatial correlation between identified system components and a series of neurotransmitter receptor/transporter maps from a PET tracer dataset³¹. This dataset includes mean binding maps from 36 PET studies, including 19 receptors and transporters across nine different neurotransmitter systems from a total of approximately 1,200 healthy individuals (Supplementary Table 6). Multiple maps from independent studies were available for a subset of receptors/transporters. Neurotransmitter receptor/transporter maps were resampled to the same space as the system component maps for the purpose of assessing spatial similarity. A Pearson's correlation across voxels was calculated as a measure of spatial similarity between system component maps and each of the 36 neurotransmitter receptor/transporter maps. To ensure the reliability of spatial correlation, we performed bootstrap analysis within both Study 1 and Study 2, resampling individual participants' contrast maps within each study and re-generating the four consensus system components (across both studies) based on 100 bootstrap samples for each study. Spatial correlations were computed between the bootstrapped consensus map and each PET tracer neurotransmitter map. The mean spatial correlations and their standard deviations across bootstrap samples (an estimate of the standard error of the correlation) are reported in radar plots and results.

This analysis does not assess error variability in the neurochemical maps themselves; therefore, we adopted a strategy of assessing replications of neurotransmitter–emotion regulation map associations across individual PET studies, which were available for many neurotransmitters (CB1, 5HT1a/1b/2a, 5HTT, VACHT, NET, mu-opioids, mGluR5, GABA, D1, D2 and DAT), and we interpreted only associations that were replicated across PET studies. Agreement across PET studies appeared to be high in most cases, although there are discrepancies (the same system can be assessed with different tracers, with sometimes divergent results).

Statistical analysis and data exclusion statement

All statistical tests in the present work are two-sided. No data points were excluded. No more than five participants' portions of the behavioral data are missing (Supplementary Figs. 2, 6 and 9).

Inclusion and ethics statement

Both Study 1 and Study 2 are registries of bio-behavioral correlates of midlife health and aging. Participants in both registries provided informed consent. The University of Pittsburgh Human Research Protection Office granted approval for Study 1 (Protocol ID: 07040037) and Study 2 (Protocol ID: 07110287) as well as their aggregation to create a common data registry (Protocol ID: 19030174). Participants were compensated \$350.00 in Study 1 and \$175.00 in Study 2 for completing all protocol visits.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The single-subject univariate beta images for both datasets, which are sufficient to reproduce the main results, are available at NeuroVault (<https://neurovault.org/collections/16266/>). Persistent identifier: <https://identifiers.org/neurovault.collection:16266>. The Neurosynth dataset used in the present study is compiled on GitHub and can be found at https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/neurosynth. The neurotransmitter receptor/transporter maps are also compiled on GitHub and are available at https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Atlases_and_parcellations/2022_Hansen_PET_tracer_maps. The system component maps are available for visualization and further analysis at

https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Individual_study_maps/2024_Bo_EmotionRegulation_BayesFactor.

Code availability

MATLAB codes and toolboxes for analyses are available at <https://github.com/canlab>. Customized codes for Bayes factor and corresponding analysis are available at https://github.com/KeBo2018/KeBo2023_EmotionReg_BayesFactor.

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Author contributions

Conceptualization: K.B. and T.D.W. Methodology: K.B., M.K., M.S. and T.D.W. Formal analysis: K.B. and T.D.W. Investigation: K.B., P.J.G. and T.D.W. Participant recruitment: T.E.K. and P.J.G. Resources: P.J.G. and T.D.W. Data pre-processing: K.B., T.E.K. and P.J.G. Writing—original draft: K.B. and T.D.W. Writing—review and editing: K.B., T.E.K., M.K., M.S., P.J.G. and T.D.W. Funding support: P.J.G. and T.D.W.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Data collection	No software was used in fmri data collection. Behavioral rating data is collected using E-Prime software (Psychology Software Tools, Inc., Sharpsburg, PA)
Data analysis	MATLAB code (Matlab R2022a) and toolbox for analyses is available at https://github.com/canlab . Customized codes for Bayes factor and corresponding analysis are available at https://github.com/KeBo2018/KeBo2023_EmotionReg_BayesFactor . DeepMRye toolbox is used for Figure S12.

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collections/16266/). Persistent identifier: <https://identifiers.org/neurovault.collection:16266>. The Neurosynth dataset used in the current study is compiled on GitHub and can be found at (https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/neurosynth). Additionally, the neurotransmitter receptor/transporter maps are also compiled on GitHub and are available at (https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Atlases_and_parcellations/2022_Hansen_PET_tracer_maps)

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We confirm that we use the terms sex and gender in ways consistent with their definitions.

The study was designed to assess participants of male and female sex, as determined by self-report. It was not powered to systematically test effects of different genders to the degree that they diverge from self-reported sex. We include self-reported sex in the metadata shared with the publication. The sex effects on main result is reported in main text and supplementary document.

Reporting on race, ethnicity, or other socially relevant groupings

The participants in both study are primarily Caucasian-American, following the population demographics of the Western Pennsylvania geographical area from which the sample was recruited, and the study was not powered to investigate differences among different racial or ethnic groups.

Population characteristics

One hundred eighty-two participants (95 males, 87 females, mean age = 43.45±7.26) from 'Study1' and one hundred seventy-six participants (87 males, 89 females, mean age = 40.58±6.33) from 'Study 2' are included in the current study. Common exclusions for participation in study 1 and study 2 were any self-reported history of a diagnosed cardiovascular disease or clinical treatment for a cardiovascular disease; self-reports of a psychotic disorder, such as schizophrenia; chronic renal or hepatic diseases; history of seizure or cerebrovascular disorders; ongoing treatment for cancer or its treatment within the year prior to enrollment; chronic respiratory diseases; and, for women, pregnancy or recent pregnancy. Other shared exclusions were (a) current usage of medications to control insulin, glucose, glucocorticoids, arrhythmias, blood pressure, lipids, weight, and mood at the time of screening; and (b) contraindications for brain imaging by magnetic resonance imaging (MRI; e.g., claustrophobia, metallic implants, etc.). Exclusion criteria that were unique to study 1 included: resting blood pressure equaling or exceeding 160mmHg systolic and/or 100mmHg diastolic blood pressure (BP); consuming 5 or more alcoholic beverages 3+ times per week; taking supplements to increase fatty acid levels; routine shift-work; and less than part time employment outside of the home (< 25 hr per week). Exclusion criteria that were unique to study 2 included: red-green color-blindness and having a resting blood pressure equaling or exceeding 140mmHg SBP and/or 90mmHg DBP.

Recruitment

Recruitment for study1 and study2 entailed mass-mailings of study advertisements to residents of the greater Allegheny County area of Pennsylvania, USA. Addresses for mailings were derived from public domain lists (e.g., voter and vehicle registration registrations, etc.) between the years 2008 to 2011 for AHAB-2 and between 2008 to 2014 for PIP. Periods of data collection thus spanned from 2008 to 2014.

Ethics oversight

Participants in study1 and study2 provided informed consent prior to testing, and study approval was granted by The University of Pittsburgh Human Research Protection Office (study1 Protocol ID: 07040037; study2 Protocol ID: 07110287). Approval was also granted to merge study1 and study2 data (Protocol ID: 19030174).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

N=182 for study 1 and N=176 for study 2. No statistical methods were used to pre-determine sample sizes, but the sample sizes in current study are significantly larger than the average for similar tasks (average n ≈ 28.6, see Morawetz et al., 2017 meta-analysis).

Data exclusions

No data were excluded from the analysis

Replication

The main findings reflect effect that are reliable across two independent datasets with relatively large sample size (n=182 and 176)

Randomization

We pseudo-randomize the order of experimental conditions to minimize order effects.

Blinding

No blinding was necessary for the study because there were no between-subject factors in current design.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA

Magnetic resonance imaging

Experimental design

Design type	Event-related design
Design specifications	Two studies applied the same task structure. In the task, participants viewed 30 unpleasant and 15 neutral IAPS images. Specifically, participants were trained and instructed to look and attend to images or decrease their negative feeling on negative images by reappraising the affective situation of pictures. Such behaviors were instructed by a 2-s cue ('Look' or 'Decrease'). The images are presented after cue for seven seconds. After the picture presentation, participants rated their negative emotion state in four seconds on a 5-point scale panel (1=neutral and 5= extremely unpleasant), followed by a randomized rest period (1-3s). The entire task period comprised fifteen 'Look neutral' trials, fifteen 'Look negative' trials and fifteen 'Decrease negative' trials.
Behavioral performance measures	Behavioral outcome was the rating of negative emotion. After the picture presentation, participants rated their negative emotion state in four seconds on a 5-point scale panel (1=neutral and 5= extremely unpleasant),

Acquisition

Imaging type(s)	Functional
Field strength	3T
Sequence & imaging parameters	fMRI acquisition parameters: FOV = 205×205mm, matrix size = 64×64, TR = 2000ms, TE = 28ms, and FA = 90°.
Area of acquisition	Whole brain
Diffusion MRI	☐ Used ☒ Not used

Preprocessing

Preprocessing software	fMRI data were preprocessed with statistical parametric mapping software (SPM12; http://www.fil.ion.ucl.ac.uk/spm).
Normalization	Images were co-registered to each participant's skull-stripped and biased-corrected MPAGE image. Co-registered images were then normalized to Montreal Neurological Institute (MNI) space and interpolated to 2 × 2 × 2 mm voxels.
Normalization template	Montreal Neurological Institute (MNI) space

Noise and artifact removal

Prior to first-level model, High pass filter of 1/180 Hz was applied to exclude the possible low-frequency temporal drift. In addition, six head motion parameters (x, y, z, roll, pitch and yaw), spikes (Coded as 1 for outlier and 0 for other volumes) and time series of white matter and cerebrospinal fluid (CSF) are included as nuisance regressors. Time series of fMRI responses within standard white matter and CSF were extracted using canonical masks in CANlab tool (<https://github.com/canlab/CanlabCore>). Six head motion covariates were extracted from realignment parameters. Images defined as spikes if they are outside the 95% confidence regions of the cloud of images in multidimensional space. The distance in this space is referred to as mahalanobis distance. The spike number is 19 ± 7.7 for 'Study 2' and 17 ± 8.0 for 'Study 1', which represents approximately 5% of the scans.

Volume censoring

No censoring, instead inclusion of motion spikes as regressors in the first-level model

Statistical modeling & inference

Model type and settings

Mass univariate model was applied in both 1st and 2nd level analysis to calculate the voxel-level activation for brain contrast of emotion generation and reappraisal. 2nd level activations were then evaluated by Bayes factor for the purpose of subsequent axiomatic approach

Effect(s) tested

Bayes Factors (BFs) are applied to quantify and compare the support evidence for both null and alternative hypotheses for each contrast. BF value reflects the likelihood ratio between alternative and null. BF values in the current study are computed using JZS prior and the method was provided in the previous study (Rouder et al., 2009). In the method, T statistics from 2nd-level analysis are utilized to compute BF value. BF threshold is 10 for alternative and 1/10 for null hypothesis. This threshold denotes strong evidence for corresponding hypothesis (Andrzejewicz et al., 2015). If a voxel satisfied all BF and statistical requirement of a specific system, it'll be assigned in this system component.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☒ Both

Anatomical location(s)

Amygdala: Defined in Amunts et al., 2020. The PAG region was defined anatomically by T.D.W. in 2018. See https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Atlases_and_parcellations

Statistic type for inference

In both individual studies and consensus map, cluster size is controlled to be equal or larger than 15 voxels.

(See [Eklund et al. 2016](#))

Correction

FDR correction

Models & analysis

n/a	Involved in the study
☒	☐ Functional and/or effective connectivity
☒	☐ Graph analysis
☒	☐ Multivariate modeling or predictive analysis